Seventy-four Candidates for Datasets 50–100

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August 25, 2026

The database holds 49 schematized and L0–L4-verified *Saccharomyces cerevisiae* datasets. Reaching 100 needs 51 more. Table 2 ranks 74 candidates and Table 3 gives each one's citation, link and data location; rows 1–51 are the recommended set and the remaining 23 are a ranked reserve. Table 6 records what was considered and dropped, which is where the one hard constraint shows up: a dataset earns a place only if the total genomic content of each strain can be reconstructed.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1 The selection rule ✗

1.1 Sequence reconstruction is the gate, not a preference ✗

A phenotype is only trainable against a genotype that can be written down. Every row in Table 2 therefore names a **sequence basis**: the route by which the total genomic content of one strain would be reconstructed. The nine bases in use are

- **S288C-KO** – the reference genome minus one cataloged ORF (BY4741, BY4742, BY4743). Twenty-one of the recommended 51 sit here.
- **S288C-KO/het** – a heterozygous diploid deletion, so a dosage edit rather than a null.
- **S288C+guide** – the genome is unedited and the perturbation is a designed cassette plus a guide target. Regulatory, not sequence-level.
- **S288C+designed-edit** – a designed single-nucleotide or small-indel library. Designed, and not verified per strain.
- **S288C+tag** – the reference plus a designed tag, degron or promoter cassette.
- **S288C+ORF-plasmid** – the reference plus a barcoded ORF on a known plasmid.
- **isolate-WGS** – a published assembly or variant call set per isolate.
- **segregant-WGS** – genotype calls or whole-genome sequence per progeny.
- **engineered-chassis** – a named production strain plus heterologous cassettes.
- **reference-only** – wild type, where the environment carries the perturbation.

What fails the gate is a strain whose genome is unknown even in principle. Genome shuffling and adaptive laboratory evolution without resequencing both land there: the selected progeny differ from their parents at unmapped positions, so there is no genotype to map a phenotype from. Those are dropped in Table 6 rather than ranked low, since no amount of phenotypic value substitutes for a missing genotype.

The **S288C+designed-edit** and **S288C+guide** bases carry a weaker claim than the others and the distinction is worth holding. A designed variant library states what each member was built to be, not what it turned out to be, so its genotypes are designed rather than realized. Ingesting one commits to storing the design and a verification status, not a verified sequence.

1.2 Tier, and what rows are ranked on ✗

Tier is assigned from stated bars rather than judgment:

- **Tier 1** clears all three of $\geq 10^3$ genotypes, $\geq 10^4$ instances, and a clean sequence basis, and lands in one of the six requested classes.
- **Tier 2** clears two of the three, or is the only dataset covering a requested class, or directly de-risks the Perturb-seq proposal.
- **Tier 3** is a modality or reference backbone the substrate needs without itself carrying a large genotype axis.
- **Tier 4** is real but small, coarsely labeled, or expensive to extract.

The bars apply to what a row contains rather than to its reputation. Piotrowski 2017 has the largest compound library in yeast at 13,524 compounds and still fails Tier 1, because 157 strains is not a genotype axis. Turco 2023 has the widest environment axis at 7,536 and fails for a different reason: it is a meta-aggregation, so its instance count re-serves primary screens, several of them already built, and the net-new fraction is unknown until it is de-duplicated against the rest of this table.

Within a tier, rows sort by **measurements**, meaning instances times phenotype dimensionality. Instances alone is the wrong statistic and gets the order visibly wrong: Muenzner 2024 is 796 instances, which ranks it below a mid-sized fitness screen, but each instance is a proteome, so it is roughly 1.6×10^6 numbers. This is the same normalization the supported-datasets table already applies through its gzip-signal column.

1.3 Where the cut falls, and what it cost ✗

The cut at 51 is arithmetic, $100 - 49$, not a quality threshold. Rows on both sides of it are close, so it should be read as a budget rather than a verdict. One row was pinned into the recommended set because it was named explicitly in the scoping request rather than because it ranked there: Table 1 names the pin and the row it displaced, so the recommended set stays auditable.

Table 1: Rows pinned into the recommended set because they were named explicitly in the scoping request, and the row each displaced. Displacement picks the weakest non-requested row by tier first and measurement count second, so a pin costs the least it can and cannot evict a tier-1 row to make room for a lower one.

Pinned in	Displaced to the reserve
Trikka 2015 (carotenogenic heterozygous screen)	Chang 2012 (organic-acid production screen)

Displacement picks the weakest non-requested row by tier first and measurements second. The tier term is load-bearing rather than decorative: ranking on measurements alone selected Puddu 2019 for eviction, and that row is the whole-collection sequencing on which every **S288C-KO** sequence basis in this table rests.

2 The ranked list ✗

Table 2: Ranked candidates to take the database from 49 supported datasets to 100. *Genotypes* and *Env* are the perturbation and condition axes. *Instances* is the number of genotype×environment records, marked † where it is the product of the two axes rather than a figure the paper reports, and ‡ where it is an order-of-magnitude estimate. *Meas.* is instances times phenotype dimensionality, which is what a vector-valued panel actually contributes and what rows are ranked on. *Sequence basis* is how the total genomic content of one strain would be reconstructed; a row with no such route is excluded rather than ranked (Table 6). Tier is defined in Sec. 1. A bullet marks a row that bears directly on the Perturb-seq proposal (Table 5). *Link* resolves to the source and is clickable; the full citation and the data location are in Table 3. A superscript **B** marks a row already attempted and blocked on data access, and **L** one that already has a loader in flight; neither is an untouched candidate. Rows 1–51 are the recommended set.

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
1	**Lee 2014 (HIP-HOP fitness signatures)** [B]	Tolerance / robustness	11,000 (het + hom)	3,356 compounds	1.3×10^7	1.3×10^7	chemogenomic fitness score	S288C-KO	doi:10.1126/science.1250217
2	**Jakobson 2025 (genome-to-proteome map)**	Natural variation	800 F6 segregants	1	800	1.6×10^6	protein abundance, >6,400 pQTL	segregant-WGS	doi:10.1126/science.adu3198
3	**Muenzner 2024 (natural-isolate proteome)**	Natural variation	796 natural isolates	1	796	1.6×10^6	protein abundance (DIA-MS)	isolate-WGS	doi:10.1038/s41586-024-07442-9
4	**Hale 2024 (CRISPRi x natural variation)** •	Natural variation	169 segregants x 1,721 genes	2 induced	1.4×10^6†	1.4×10^6	relative fitness (double-barcode seq)	segregant-WGS	doi:10.1038/s41467-024-48626-1
5	**Galardini 2019 (four backgrounds x 38 conditions)**	Natural variation	3,786 KOs x 4 backgrounds	38 conditions	5.8×10^5†	5.8×10^5	colony-size fitness (S-score)	isolate-WGS	doi:10.15252/msb.20198831
6	**Bloom 2019 (16-parent cross)**	Natural variation	14,000 segregants	38 traits	5.3×10^5†	5.3×10^5	quantitative growth traits	segregant-WGS	doi:10.7554/eLife.49212
7	**Parsons 2006 (bioactive-compound profiling)**	Tolerance / robustness	5,000 viable YKO	82 compounds	4.1×10^5†	4.1×10^5	hypersensitivity score	S288C-KO	doi:10.1016/j.cell.2006.06.040
8	**Dutta 2026 (barcoded natural-isolate chemical response)**	Natural variation	520 natural isolates	>600 compounds	3.1×10^5†	3.1×10^5	pooled-barcode fitness	isolate-WGS	doi:10.1038/s41467-026-73532-z
9	**Cooper 2010 (CE-MS amino-acid metabolome)**	Metabolite / precursor	4,700 YKO	1	4.7×10^3	9.4×10^4	free amino-acid pools (capillary electrophoresis)	S288C-KO	doi:10.1101/gr.105825.110
10	**Aulakh 2025 (genome-scale ionome)**	Metabolite / precursor	4,800 YKO	1	4.8×10^3	4.8×10^4	intracellular metal-ion content (ICP-MS)	S288C-KO	doi:10.1016/j.cels.2025.101319
11	**Peter 2018 (1,011 isolate genomes + phenome)**	Natural variation	1,011 isolates (971 phenotyped)	36 conditions	3.5×10^4†	3.5×10^4	growth fitness per condition	isolate-WGS	doi:10.1038/s41586-018-0030-5
12	**Zhang 2021 (YETI titratable overexpression)**	Metabolite / precursor	1,022 essential + 4,668 non-essential	beta-estradiol dose series	2.3×10^4‡	2.3×10^4	fitness across an induction gradient	S288C+tag	doi:10.15252/msb.202010321
13	**Peeters 2021 (fermentation-trait QTL atlas)**	Natural variation	1,125 sequenced segregants	18 traits	2.0×10^4†	2.0×10^4	ethanol / glycerol / isobutanol titer, stress resistance, aroma	segregant-WGS	PMID 34727964
14	**Van Leeuwen 2024 (short-chain organic acids)**	Tolerance / robustness	4,800 YKO	acetic, butyric, octanoic	1.4×10^4†	1.4×10^4	fitness per acid	S288C-KO	PMC10903034

Table 2, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
15	**Chica 2026 (AutoDRY autophagy screen)**	Modality / backbone	4,760 KO + 1,159 DAmP	2 (nitrogen starvation, replete)	1.2×10^4†	1.2×10^4	autophagy flux (deep-learning image classifier)	S288C-KO	doi:10.1038/s41556-025-01837-0
16	**Sopko 2006 (genome-scale overexpression toxicity)**	Metabolite / precursor	5,280 ORFs	GAL induced vs glucose	1.1×10^4†	1.1×10^4	growth-rate reduction on induction	S288C+ORF-plasmid	doi:10.1016/j.molcel.2005.12.011
17	**Liu 2021 (tryptophan / isobutanol tolerance)**	Tolerance / robustness	5,006 YKO	+/- isobutanol	1.0×10^4†	1.0×10^4	colony-size fitness (ScreenMill imaging)	S288C-KO	doi:10.1186/s13068-021-02048-z
18	**Kuroda 2019 (isobutanol-specific tolerance)**	Tolerance / robustness	4,800 YKO	isobutanol vs ethanol	9.6×10^3†	9.6×10^3	isobutanol-specific fitness	S288C-KO	doi:10.1016/j.cels.2019.10.006
19	**Puddu 2019 (WGS of the deletion collection)**	Modality / backbone	4,800 deletion strains	1	4.8×10^3	4.8×10^3	structural variants, CNV, aneuploidy	S288C-KO	doi:10.1038/s41586-019-1549-9
20	**Turco 2023 (Yeast Phenome)**[L]	Tolerance / robustness	5,000	7,536 environments	3.8×10^7†	3.8×10^7	growth / fitness per environment	S288C-KO	doi:10.1126/sciadv.adg5702
21	**N'Guessan 2025 (segregant scRNA-seq eQTL)** •	Expression / single cell	4,500 profiled segregants	1	4.5×10^3	2.7×10^7	single-cell transcriptome + eQTL	segregant-WGS	PMC12303567
22	**Piotrowski 2017 (MOSAIC diagnostic panel)**	Tolerance / robustness	157 diagnostic	13,524 compounds	2.1×10^6†	2.1×10^6	chemogenomic fitness score	S288C-KO	doi:10.1038/nchembio.2436
23	**Gasch 2000 (environmental stress response)**	Expression / single cell	wild type	9 stress series, multi-timepoint	150‡	9.0×10^5	mRNA abundance time course	reference-only	doi:10.1091/mbc.11.12.4241
24	**Decourty 2021 (RNA-metabolism genetic interactions)**	Modality / backbone	700,000 double mutants	1	7.0×10^5	7.0×10^5	genetic-interaction score	S288C-KO	doi:10.1093/nar/gkab680
25	**Leutert 2023 (phosphoproteome x 101 conditions)**	Expression / single cell	wild type	101 conditions	101	5.0×10^5	phosphosite abundance	reference-only	doi:10.1038/s41594-023-01115-3
26	**Wildenhain 2016 (chemical-genetic matrix, Sci Data)**	Tolerance / robustness	242 diagnostic	5,518 compounds	4.9×10^5	4.9×10^5	chemical-genetic interaction score	S288C-KO	doi:10.1038/sdata.2016.95
27	**Chong 2015 / CYCLoPs (single-cell proteome dynamics)** •	Modality / backbone	4,100 GFP-tagged ORFs	cell-cycle / stress	2.5×10^4‡	4.2×10^5	abundance + localization class	S288C+tag	doi:10.1016/j.cell.2015.04.051
28	**Dong 2021 (MAGIC + SAM biosensor)**	CRISPR library screen	100,000 guides (a/i/d)	1	1.0×10^5‡	1.0×10^5	SAM biosensor fluorescence (FACS)	S288C+guide	doi:10.1016/j.ymben.2021.03.005
29	**Momen-Roknabadi 2020 (inducible CRISPRi library)** •	CRISPR library screen	genome-wide guide library	nutrient limitations	9.0×10^4‡	9.0×10^4	per-guide fitness under induction	S288C+guide	doi:10.1038/s42003-020-01452-9
30	**Hackett 2016 (SIMMER multi-omic flux)**	Expression / single cell	wild type	25 chemostat states	25	7.5×10^4	paired metabolome + proteome + flux	reference-only	doi:10.1126/science.aaf2786

Table 2, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
31	**McGlincy 2021 (genome-scale CRISPRi library)** •	CRISPR library screen	61,094 guides, 10/gene	continuous culture	6.1×10^4	6.1×10^4	per-guide fitness	S288C+guide	doi:10.1186/s12864-021-07518-0
32	**Bao 2018 (CHAnGE single-nucleotide library)**	CRISPR library screen	tens of thousands of designed variants	inhibitor selection	6.0×10^4‡	6.0×10^4	variant fitness (pooled barcode seq)	S288C+designed-edit	doi:10.1038/nbt.4132
33	**Yoshikawa 2011 (deletion + overexpression phenome)**	Tolerance / robustness	4,800 KO + matched overexpression	carbon sources / stress	5.8×10^4‡	5.8×10^4	growth-rate fitness	S288C-KO	doi:10.1534/g3.111.000695
34	**Ho 2009 (MoBY-ORF barcoded overexpression)**	Metabolite / precursor	5,100 barcoded ORFs	compound panel	5.1×10^4‡	5.1×10^4	fitness per compound	S288C+ORF-plasmid	doi:10.1038/nbt.1534
35	**Zhu 2014 (kinase / phosphatase lipidomics)**	Metabolite / precursor	129 kinase + phosphatase KOs	1	129	2.6×10^4	lipid class and species abundance	S288C-KO	PMC4196872
36	**Xiao 2014 (genome-wide RNAi furfural tolerance)**	Tolerance / robustness	genome-scale RNAi library	furfural vs control	2.0×10^4‡	2.0×10^4	knockdown enrichment under furfural	S288C+guide	doi:10.1186/1754-6834-7-78
37	**Crook 2016 (tunable RNAi, isobutanol + 1-butanol)**	Tolerance / robustness	genome-scale tunable RNAi	isobutanol, 1-butanol	2.0×10^4‡	2.0×10^4	dose-graded knockdown fitness under alcohol stress	S288C+guide	PMID 27654654
38	**Douglas 2012 (barcoded overexpression fitness)**	Metabolite / precursor	genome-scale barcoded ORFs	baseline + stress	2.0×10^4‡	2.0×10^4	pooled competitive fitness on induction	S288C+ORF-plasmid	doi:10.1534/g3.112.003400
39	**Costello 2020 (bioreactor Bar-seq)**	Tolerance / robustness	pooled YKO library	feed schemes / pH	1.9×10^4‡	1.9×10^4	fitness under fed-batch cultivation	S288C-KO	doi:10.1186/s12934-020-01423-z
40	**Roy 2018 (multiplexed precision editing)**	CRISPR library screen	thousands of designed edits	1	1.6×10^4‡	1.6×10^4	variant fitness	S288C+designed-edit	doi:10.1038/nbt.4137
41	**Gerke 2017 (urea-cycle mQTL)**	Natural variation	147 diploid segregants	1	147	1.5×10^4	untargeted metabolite abundance	segregant-WGS	doi:10.1534/genetics.117.201107
42	**Gurvitz (fatty-acid utilization screen)**	Metabolite / precursor	4,800 viable YKO	oleate, myristate, acetate	1.4×10^4†	1.4×10^4	fatty-acid utilization competence	S288C-KO	embopress.org/doi/pdf/10.1038/msb4100051
43	**Gameiro 2025 (AID2 degron libraries)**	Modality / backbone	>5,600 tagged ORFs	baseline + hydroxyurea	1.1×10^4†	1.1×10^4	degradation efficiency; fitness under stress	S288C+tag	doi:10.1083/jcb.202409007
44	**Si 2017 (RAGE RNAi, isobutanol titer)**	CRISPR library screen	genome-scale RNAi, iterative rounds	isobutanol selection	1.0×10^4‡	1.0×10^4	construct enrichment; isobutanol titer for top strains	S288C+guide	doi:10.1038/ncomms15187
45	**HamediRad 2018 (RAGE xylose utilization)**	CRISPR library screen	genome-scale RNAi	xylose-limited	1.0×10^4‡	1.0×10^4	enrichment; xylose consumption rate	S288C+guide	doi:10.1002/bit.26570
46	**Pereira 2014 (wheat-straw hydrolysate)**	Tolerance / robustness	EUROSCARF deletion collection	hydrolysate vs control	9.6×10^3†	9.6×10^3	tolerance to industrial hydrolysate	S288C-KO	doi:10.1007/s10295-014-1519-z
47	**Endo 2008 (vanillin tolerance)**	Tolerance / robustness	diploid deletion collection	vanillin vs control	9.6×10^3†	9.6×10^3	vanillin sensitivity	S288C-KO	doi:10.1128/AEM.01541-08

Table 2, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
48	**Sharifpoor 2012 (kinome synthetic dosage lethality)**	Modality / backbone	92 kinase queries, genome-wide array	1	8.7×10^3	8.7×10^3	genetic interaction / SDL score	S288C-KO	doi:10.1101/gr.129213.111
49	**Ambroset 2014 (metabolite QTL)**	Natural variation	100 segregants	1	100	7.4×10^3	74 metabolite concentrations (LC-MS/MS)	segregant-WGS	doi:10.1371/journal.pgen.1004142
50	**Blank 2005 (13C metabolic flux)**	Metabolite / precursor	200 viable null mutants	glucose minimal	200‡	6.0×10^3	intracellular flux distribution (13C-MFA)	S288C-KO	doi:10.1186/gb-2005-6-6-r49
51	**Trikka 2015 (carotenogenic heterozygous screen)**	Metabolite / precursor	4,700 heterozygous deletions	1	4.7×10^3	4.7×10^3	carotenoid color; sclareol titer for top strains	S288C-KO/het	doi:10.1186/s12934-015-0246-0
	Cut line. Rows above are the 51 that reach 100; rows below are the ranked reserve.								
52	**Chang 2012 (organic-acid production screen)**	Metabolite / precursor	4,800 YKO	1	4.8×10^3	4.8×10^3	acetate / pyruvate / succinate halo (pH indicator)	S288C-KO	PMID 22277779
53	**Mukherjee 2021 (CRISPRi essential genes x acetic acid)**	Tolerance / robustness	1,100 essential genes	acetic acid vs control	2.2×10^3†	2.2×10^3	fitness under acid stress	S288C+guide	doi:10.1128/mSystems.00410-21
54	**Breslow 2008 (DAmP hypomorphic collection)**	Modality / backbone	842 haploid + 970 diploid essential	1	1.8×10^3	1.8×10^3	competitive fitness	S288C+tag	doi:10.1038/nmeth.1234
55	**Kofoed 2015 (barcoded ts alleles)**	Modality / backbone	600+ barcoded ts alleles	permissive / restrictive	1.2×10^3†	1.2×10^3	ts severity (growth ratio)	S288C+designed-edit	doi:10.1534/g3.115.019174
56	**Lian 2017 (CRISPR-AID, beta-carotene)**	CRISPR library screen	combinatorial a/i/d triples	1	200‡	200	beta-carotene titer	engineered-chassis	doi:10.1038/s41467-017-01695-x
57	**Cubillos 2017 (nitrogen-consumption QTL)**	Natural variation	165 sequenced F12 segregants	nitrogen-limited fermentation	165	165	nitrogen consumption	segregant-WGS	doi:10.1534/g3.117.042127
58	**Braberg 2020 (point-mutant E-MAP)**	Modality / backbone	350 point mutants x genome-wide array	1	5.0×10^5	5.0×10^5	residue-level genetic interaction	S288C+designed-edit	doi:10.1126/science.aaz4910
59	**Weill 2018 (SWAT libraries)**	Modality / backbone	5,500 acceptor-tagged strains	1	5.5×10^3	1.6×10^4	abundance, localization, topology	S288C+tag	doi:10.1038/s41592-018-0044-9
60	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)** •	Expression / single cell	clonal wild type	baseline + stress	2‡	1.2×10^4	single-cell transcriptome	reference-only	doi:10.1038/s41564-018-0346-9
61	**Gasch 2017 (single-cell stress heterogeneity)** •	Expression / single cell	wild type	stress vs unstressed	2‡	1.2×10^4	single-cell transcriptome	reference-only	doi:10.1371/journal.pbio.2004050

Table 2, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
62	**Valenti 2025 (SWAT degron + GFP collection)**	Modality / backbone	proteome-wide SWAT library	depleted / not	1.0×10^4‡	1.0×10^4	depletion efficiency + abundance/localization	S288C+tag	doi:10.1083/jcb.202409050
63	**Ghaemmaghami 2003 (absolute protein abundance)**	Modality / backbone	4,251 TAP-tagged ORFs	1	4.3×10^3	4.3×10^3	absolute abundance (molecules/cell)	S288C+tag	doi:10.1038/nature02046
64	**Huh 2003 (GFP localization atlas)**	Modality / backbone	4,156 GFP-tagged ORFs	1	4.2×10^3	4.2×10^3	subcellular localization (22 classes)	S288C+tag	doi:10.1038/nature02026
65	**Michaelis 2023 (protein interactome)**	Modality / backbone	proteome-scale AP-MS baits	1	4.0×10^3‡	4.0×10^3	protein-protein interaction + interface	S288C+tag	doi:10.1038/s41586-023-06739-5
66	**Tengolics 2024 (domestication metabolome)**	Natural variation	17 S. cerevisiae populations	1	17	1.6×10^3	19 amino acids + 78 other metabolites	isolate-WGS	doi:10.1073/pnas.2313354121
67	**Schulte 2023 (mitochondrial complexome)**	Modality / backbone	900 mitochondrial proteins	1	900	900	complex assembly profile	reference-only	doi:10.1038/s41586-022-05641-w
68	**Eder 2020 (flux QTL)**	Natural variation	segregant panel (n unconfirmed)	wine fermentation	100‡	100	modeled intracellular flux	segregant-WGS	PMID 32034164
69	**Yue 2017 (wild vs domesticated assemblies)**	Natural variation	12 long-read assemblies	n/a	12	12	genome structure, TE and subtelomere dynamics	isolate-WGS	doi:10.1038/ng.3847
70	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq)** •	Expression / single cell	method	method	–‡	–	single-cell transcriptome	reference-only	doi:10.1002/yea.3927
71	**Mulleder 2012 (prototrophic deletion collection)**	Modality / backbone	5,185 prototrophic + 839 titratable essential	1	–‡	–	strain resource (no phenotype)	S288C-KO	doi:10.1038/nbt.2442
72	**Zackrisson 2016 (Scan-o-matic growth curves)**	Modality / backbone	YKO-compatible platform	panel-dependent	–‡	–	full growth curve (rate, lag, yield)	S288C-KO	doi:10.1534/g3.116.032342
73	**Bennett 2001 (ionizing-radiation resistance)**	Tolerance / robustness	4,800 YKO	irradiated vs control	9.6×10^3†	9.6×10^3	radiation sensitivity	S288C-KO	doi:10.1038/ng778
74	**Sugiyama (lactic-acid tolerance)**	Tolerance / robustness	4,800 YKO	4 percent lactic acid vs control	9.6×10^3†	9.6×10^3	acid tolerance	S288C-KO	nisr.or.jp/wp-content/uploads/NISR06sugiyama.pdf

Table 3: Sources for Table 2, in the same order. *Why* states what the row buys that the built set does not. *Data* is where the per-record values live; entries marked unconfirmed were not fetched live and must be checked before a loader is written. Every link is clickable.

#	Dataset, citation and link	Why	Data
1	**Lee 2014 (HIP-HOP fitness signatures)** Lee AY, St Onge RP, Proctor MJ, et al., Giaever G, Nislow C. Science 2014;344:208-211. doi:10.1126/science.1250217	Largest chemical-genetic matrix in existence, on the exact YKO genotype axis Costanzo and Kuzmin already use. Biggest single instance-count gain available. BLOCKED: WS15 already attempted it and it stands as awaiting author matrices, so the Science SI and the lab portal did not yield per-strain values. Unblocking is an author request, not a loader.	Science SI + Nislow/Giaever portal, neither of which yielded per-strain matrices; awaiting author matrices per the WS15 roadmap
2	**Jakobson 2025 (genome-to-proteome map)** Jakobson CM, et al. Science 2025;390:eadu3198. doi:10.1126/science.adu3198	Natural alleles mapped to enzyme abundance. A lower-risk engineering lever than a heterologous construct, and the proteome counterpart to the metabolite-QTL panels.	Science data-availability statement (accession unconfirmed)
3	**Muenzner 2024 (natural-isolate proteome)** Muenzner J, et al., Ralser M. Nature 2024;630:149-157. doi:10.1038/s41586-024-07442-9	Proteomes on the sequenced isolate panel whose transcriptomes Caudal 2024 already supplies. That overlap is the paired anchor the RNA-to-protein inference thesis needs.	PRIDE PXD048219
4	**Hale 2024 (CRISPRi x natural variation)** Hale JJ, Matsui T, Goldstein I, et al., Kruglyak L. Nat Commun 2024;15:4234. doi:10.1038/s41467-024-48626-1	Directly measures how a genetic perturbation's effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.	SRA PRJNA986287 + Figshare
5	**Galardini 2019 (four backgrounds x 38 conditions)** Galardini M, Busby BP, Vieitez C, Dunham AS, Typas A, Beltrao P. Mol Syst Biol 2019;15:e8831. doi:10.15252/msb.20198831	The same knockout built independently in four sequenced backgrounds. Quantifies the 18.5 percent of deletion phenotypes that do not transfer, which is the generalization risk the whole substrate is exposed to.	GEO GSE123118 + github.com/mgalardini/2018koyeast
6	**Bloom 2019 (16-parent cross)** Bloom JS, Boocock J, Treusch S, Sadhu MJ, Day L, Oates-Barker H, Kruglyak L. eLife 2019;8:e49212. doi:10.7554/eLife.49212	Largest sequenced recombinant panel in yeast. Sixteen founders means allelic diversity a two-parent cross cannot reach, and the genotypes are sequence, not markers.	eLife SI + SRA (confirm accession before ingestion)
7	**Parsons 2006 (bioactive-compound profiling)** Parsons AB, Lopez A, Givoni IE, et al., Boone C. Cell 2006;126:611-625. doi:10.1016/j.cell.2006.06.040	Full-collection strain coverage against a modest compound panel, the complement to the diagnostic-panel screens above.	Cell SI tables (accession unconfirmed)
8	**Dutta 2026 (barcoded natural-isolate chemical response)** Dutta A, Garin M, Loegler V, Brach G, Friedrich A, Yoshimura M, Hirano H, Osada H, Boone C, Yashiroda Y, Hou J, Schacherer J. Nat Commun 2026. doi:10.1038/s41467-026-73532-z	Natural sequence diversity crossed with a chemogenomic panel, on isolates drawn from the sequenced 1,011 collection. The same strains already carry Caudal 2024 transcriptomes and Muenzner 2024 proteomes, so it is a three-modality join on one genotype axis.	Nat Commun SI; isolates from the 1,011 panel (ENA)
9	**Cooper 2010 (CE-MS amino-acid metabolome)** Cooper SJ, Finney GL, Brown SL, Nelson SI, Hesse J, MacCoss MJ, Fields S. Genome Res 2010;20:1288-1296. doi:10.1101/gr.105825.110	An independent platform measuring the same trait class as the already-built Mulleder 2016 on an overlapping genotype axis. Two independent measurements of one phenotype is the cleanest available test of whether a model has learned biology or a batch.	Genome Research SI tables
10	**Aulakh 2025 (genome-scale ionome)** Aulakh SK, et al. Cell Syst 2025;16:101319. doi:10.1016/j.cels.2025.101319	The only gene-indexed ionome. Metal-cofactor supply gates iron-sulfur and zinc-dependent pathway enzymes, and no built dataset measures it.	Cell Systems SI (accession unconfirmed)

Table 3, continued

#	Dataset, citation and link	Why	Data
11	**Peter 2018 (1,011 isolate genomes + phenome)** Peter J, De Chiara M, Friedrich A, et al., Schacherer J. Nature 2018;556:339-344. doi:10.1038/s41586-018-0030-5	The genome backbone every other natural-variation row resolves against, and it carries its own phenotype panel. 1,625,809 high-quality SNPs at 232-fold mean coverage.	ENA/SRA; phenotype tables via the paper and France Genomique
12	**Zhang 2021 (YETI titratable overexpression)** Zhang Y, Ho Yee Chow M, Ling Y, et al., Andrews BJ. Mol Syst Biol 2021;17:e10321. doi:10.15252/msb.202010321	Native-locus, continuously titratable overexpression. The dosage axis pathway engineering actually uses, and the only gain-of-function library that is not binary.	Mol Syst Biol SI (repository accession unconfirmed)
13	**Peeters 2021 (fermentation-trait QTL atlas)** Peeters B, Reijbroek F, Verbaet J, et al., Jarosz DF, Verstrepen KJ. 2021. PMID 34727964. PMID 34727964	The only large sequenced panel with direct isobutanol, ethanol and glycerol titers. Hits the natural-isolate and the isobutanol asks at once.	Journal SI; raw reads at SRA/ENA (accession unconfirmed)
14	**Van Leeuwen 2024 (short-chain organic acids)** Shared and specific genetic determinants of tolerance to acetic, butyric and octanoic acid. PMC10903034. PMC10903034	Three chemically homologous acids on one strain panel, so shared and acid-specific mechanisms separate within a single study rather than across incomparable screens.	Journal SI (open-access PMC)
15	**Chica 2026 (AutoDRY autophagy screen)** Chica N, et al. Nat Cell Biol 2026;28:465-479. doi:10.1038/s41556-025-01837-0	Largest single-condition imaging screen with a live Dryad deposit, and it covers essential-adjacent genes via DAmP rather than stopping at the non-essential set.	Dryad 10.5061/dryad.cfxpnvxdh
16	**Sopko 2006 (genome-scale overexpression toxicity)** Sopko R, Huang D, Preston N, et al., Boone C, Andrews B. Mol Cell 2006;21:319-330. doi:10.1016/j.molcel.2005.12.011	769 genes are growth-inhibitory when overexpressed. Overexpression toxicity is the constraint that caps titer when a pathway enzyme is pushed, and nothing in the built set measures it.	Mol Cell SI tables
17	**Liu 2021 (tryptophan / isobutanol tolerance)** Liu H-L, Wang CH-T, Chiang EP-I, Huang C-C, Li W-H. Biotechnol Biofuels 2021;14:200. doi:10.1186/s13068-021-02048-z	A second, independent full-collection isobutanol screen with a different readout. Paired with Kuroda it gives a same-phenotype, same-genotype, different-method replicate.	GEO GSE175794 (companion RNA-seq); screen in BMC SI
18	**Kuroda 2019 (isobutanol-specific tolerance)** Kuroda K, Hammer SK, Watanabe Y, Montano Lopez J, Fink GR, Stephanopoulos G, Ueda M, Avalos JL. Cell Syst 2019;9:534-547.e5. doi:10.1016/j.cels.2019.10.006	The isobutanol screen, designed against an ethanol comparator so the alcohol-general response subtracts out. Its GLN3 hit raised production 4.9-fold, a rare tolerance-to-titer translation.	ArrayExpress E-MTAB-8175 (RNA-seq); screen data in Cell Syst SI
19	**Puddu 2019 (WGS of the deletion collection)** Puddu F, et al., Jackson SP. Nature 2019;573:416-420. doi:10.1038/s41586-019-1549-9	Whole-genome sequence for every strain in the deletion collection. This is what turns the S288C-KO sequence basis from an assumption into a measurement, and it applies to roughly half the rows in this table at once.	ENA/SRA (accession unconfirmed)
20	**Turco 2023 (Yeast Phenome)** Turco G, Chang C, Wang RY, et al., Boone C, Andrews BJ, Roth FP. Sci Adv 2023;9:eadg5702. doi:10.1126/sciadv.adg5702	The widest environment axis in yeast, 96 percent chemical. Held out of tier 1 because it is a meta-aggregation: its instance count re-serves primary screens, several already built, so the net-new fraction is unknown until it is de-duplicated against the rest of this table. A loader already exists and retains 49 growth screens, so this is in flight rather than net-new work; Lee 2014 is not among those screens, so YeastPhenome does not backfill row 1.	yeastphenome.org (Zenodo 10.5281/zenodo.7714347); loader torchcell/datasets/scerevisiae/yeastphenome.py, 49 screens, not yet in the built set

Table 3, continued

#	Dataset, citation and link	Why	Data
21	**N'Guessan 2025 (segregant scRNA-seq eQTL)** N'Guessan A, Boocock J, Kruglyak L, Albert FW. eLife 2025. PMC12303567. PMC12303567	Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale.	eLife data availability; segregant panel from Bloom lineage
22	**Piotrowski 2017 (MOSAIC diagnostic panel)** Piotrowski JS, Li SC, Deshpande R, et al., Boone C, Myers CL. Nat Chem Biol 2017;13:982-993. doi:10.1038/nchembio.2436	Highest compound count of any yeast screen. Held out of tier 1 on the genotype bar: 157 strains buys condition breadth, not genotype breadth, so it trains a compound embedding rather than a genotype model.	MOSAIC portal (mosaic.cs.umn.edu) + Nat Chem Biol SI
23	**Gasch 2000 (environmental stress response)** Gasch AP, Spellman PT, Kao CM, et al., Brown PO. Mol Biol Cell 2000;11:4241-4257. doi:10.1091/mbc.11.12.4241	The canonical stress-response signature. Every genotype-indexed transcriptome in the built set needs it as the background any generic stress response is subtracted against.	SPELL /SGD Expression Connection; Stanford legacy mirror
24	**Decourty 2021 (RNA-metabolism genetic interactions)** Decourty L, et al., Saveanu C. Nucleic Acids Res 2021;49:8535-8555. doi:10.1093/nar/gkab680	Structurally identical to Costanzo and Kuzmin, so the existing loader pattern applies directly, and it extends the interaction network into RNA processing.	NAR open-access SI (accession unconfirmed)
25	**Leutert 2023 (phosphoproteome x 101 conditions)** Leutert M, et al., Villen J. Nat Struct Mol Biol 2023;30:1761-1773. doi:10.1038/s41594-023-01115-3	Post-translational enzyme control acts faster than transcription and is invisible to every other layer here. Condition axis only.	PRIDE PXD034997
26	**Wildenhain 2016 (chemical-genetic matrix, Sci Data)** Wildenhain J, Spitzer M, Dolma S, et al., Tyers M. Sci Data 2016;3:160095. doi:10.1038/sdata.2016.95	Released as an ML-ready ISA-Tab benchmark, so ingestion cost is close to zero. Distinct from the Wildenhain 2015 sentinel panel already in the table; held out of tier 1 on the same 242-strain genotype bar as Piotrowski.	ChemGRID (chemgrid.org/cgm) + PubChem BioAssay
27	**Chong 2015 / CYCLoPs (single-cell proteome dynamics)** Chong YT, Koh JLY, Friesen H, et al., Andrews BJ, Boone C. Cell 2015;161:1413-1424. doi:10.1016/j.cell.2015.04.051	Per-cell rather than per-strain-mean labels, over 20 million cells. The distributional readout a Perturb-seq design has to be scored against.	thecellvision.org/cyclops (bulk CSV)
28	**Dong 2021 (MAGIC + SAM biosensor)** Dong C, Schultz JC, Liu W, Lian J, Huang L, Xu Z, Zhao H. Metab Eng 2021;66:319-327. doi:10.1016/j.ymben.2021.03.005	MAGIC re-run with a metabolite biosensor instead of a growth selection, so the label is product concentration rather than fitness. The pattern to copy for every precursor we care about, and the sibling of the already-built Lian 2019 furfural screen.	Metab Eng SI (accession unconfirmed)
29	**Momen-Roknabadi 2020 (inducible CRISPRi library)** Momen-Roknabadi A, Oikonomou P, Zegans M, Tavazoie S. Commun Biol 2020;3:723. doi:10.1038/s42003-020-01452-9	A second CRISPRi library on a different vector and guide-design rule. Whether a model trained on one library transfers to the other is a cheap, decisive test of guide-level overfitting.	Addgene + Commun Biol SI
30	**Hackett 2016 (SIMMER multi-omic flux)** Hackett SR, Zanotelli VRT, Xu W, et al., Rabinowitz JD. Science 2016;354:aaf2786. doi:10.1126/science.aaf2786	The only jointly measured metabolome, proteome and fluxome. No genotype axis, so it is a mechanism prior rather than training data, but it is the only place the three layers are measured on the same cells.	Science SI (accession unconfirmed)
31	**McGlincy 2021 (genome-scale CRISPRi library)** McGlincy NJ, Meacham ZA, Reynaud KK, Muller R, Baum R, Ingolia NT. BMC Genomics 2021;22:205. doi:10.1186/s12864-021-07518-0	Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.	ingolia-lab.org/yeast-crispri + Addgene + BMC SI

Table 3, continued

#	Dataset, citation and link	Why	Data
32	**Bao 2018 (CHAnGE single-nucleotide library)** Bao Z, HamediRad M, Xue P, Xiao H, Tasan I, Chao R, Liang J, Zhao H. Nat Biotechnol 2018;36:505-508. doi:10.1038/nbt.4132	Single-nucleotide resolution rather than whole-gene nulls. The only genotype axis in this table finer than an ORF, and the one that matches how a real strain-design edit is specified.	Nat Biotechnol SI (raw sequencing location unconfirmed)
33	**Yoshikawa 2011 (deletion + overexpression phenome)** Yoshikawa K, Tanaka T, Furusawa C, Nagahisa K, Hirasawa T, Shimizu H. G3 2011;1:247-267. doi:10.1534/g3.111.000695	The same genes phenotyped in both loss- and gain-of-function side by side. A matched over/under pair on one platform is what a dosage-aware model needs and nothing else here provides.	G3 open-access SI
34	**Ho 2009 (MoBY-ORF barcoded overexpression)** Ho CH, Magtanong L, Barker SL, et al., Boone C. Nat Biotechnol 2009;27:369-377. doi:10.1038/nbt.1534	Native-promoter overexpression against a compound panel, giving a third dosage state alongside deletion and knockdown on the same chemogenomic readout.	moby.ccbr.utoronto.ca + Andrews lab portal
35	**Zhu 2014 (kinase / phosphatase lipidomics)** Zhu Z, Loewen CJR. Mol Biol Cell 2014. PMC4196872. PMC4196872	A gene-indexed lipidome on a regulatory gene set. Complements the already-built da Silveira 2014 lipidomics, which has no kinase axis, and reads out the malonyl-CoA sink directly.	Mol Biol Cell SI Table S4 (Excel)
36	**Xiao 2014 (genome-wide RNAi furfural tolerance)** Xiao H, Zhao H. Biotechnol Biofuels 2014;7:78. doi:10.1186/1754-6834-7-78	Furfural via a knockdown rather than a deletion axis, so it reaches essential genes and gives a dose-graded rather than binary perturbation on the flagship hydrolysate inhibitor.	BMC open-access SI
37	**Crook 2016 (tunable RNAi, isobutanol + 1-butanol)** Crook N, Sun J, Morse N, Schmitz A, Alper HS. Appl Microbiol Biotechnol 2016;100:10005-10018. PMID 27654654. PMID 27654654	Two related alcohols at graded knockdown strength. The dosage axis is the signal a binary screen throws away, and Hsp70 emerged only because of it.	AMB SI (DOI is an open verification item)
38	**Douglas 2012 (barcoded overexpression fitness)** Douglas AC, Smith AM, Sharifpoor S, et al., Andrews BJ, Boone C, Nislow C. G3 2012;2:1279-1289. doi:10.1534/g3.112.003400	The same barcode-sequencing pipeline as the already-built Hillenmeyer FitDb, run in the opposite dosage direction. Lowest harmonization cost of any overexpression row.	G3 open-access SI
39	**Costello 2020 (bioreactor Bar-seq)** Costello Z, Wehrs M, Mukhopadhyay A, et al. Microb Cell Fact 2020;19:167. doi:10.1186/s12934-020-01423-z	The only fitness measurement at bioreactor scale. Shake-flask ranking is known to diverge from fed-batch, and nothing else here tests that.	Microb Cell Fact open-access SI
40	**Roy 2018 (multiplexed precision editing)** Roy KR, Smith JD, Vonesch SC, et al., St Onge RP. Nat Biotechnol 2018;36:512-520. doi:10.1038/nbt.4137	Genomic rather than plasmid barcodes, which removes the barcode-swapping artifact that quietly corrupts pooled fitness data. The methodological complement to CHAnGE.	Nat Biotechnol SI (accession unconfirmed)
41	**Gerke 2017 (urea-cycle mQTL)** Gerke J, et al., Fay JC. Genetics 2017;206:2199. doi:10.1534/genetics.117.201107	A second mQTL panel from a different cross (oak x wine), so cross-background generalization of metabolite QTL can be tested rather than assumed.	Genetics (GSA) SI
42	**Gurvitz (fatty-acid utilization screen)** Gurvitz A, et al. Mol Syst Biol (EMBO Press). embopress.org/doi/pdf/10.1038/msb4100051	Whole-collection coverage against two fatty-acid substrates. Relevant to the lipid and malonyl-CoA branch, at the cost of manual extraction from an embedded table.	Mol Syst Biol Table I (PDF-embedded)

Table 3, continued

#	Dataset, citation and link	Why	Data
43	**Gameiro 2025 (AID2 degron libraries)** Gameiro E, Juarez-Nunez KA, Fung JJ, Shankar S, Luke B, Khmelinskii A. J Cell Biol 2025;224:e202409007. doi:10.1083/jcb.202409007	Acute depletion on a minutes-to-hours timescale, covering essential genes without the pre-adaptation a deletion strain has already done.	J Cell Biol SI; strains via IMB Mainz / EUROSCARF
44	**Si 2017 (RAGE RNAi, isobutanol titer)** Si T, Chao R, Min Y, Wu Y, Ren W, Zhao H. Nat Commun 2017;8:15187. doi:10.1038/ncomms15187	One of very few library screens whose endpoint is a titer in g/L rather than a fitness proxy, and it is iterative, so it carries combinatorial rather than single-locus effects.	Nat Commun SI (accession unconfirmed)
45	**HamediRad 2018 (RAGE xylose utilization)** HamediRad M, Lian J, Li H, Zhao H. Biotechnol Bioeng 2018;115:1552-1560. doi:10.1002/bit.26570	Pentose utilization is the other half of the recalcitrant-biomass problem, and no built dataset covers a xylose selection.	Biotechnol Bioeng SI (accession unconfirmed)
46	**Pereira 2014 (wheat-straw hydrolysate)** Pereira FB, Guimaraes PMR, Gomes DG, et al., Domingues L. J Ind Microbiol Biotechnol 2014;41:1753-1761. doi:10.1007/s10295-014-1519-z	Real industrial wheat-straw hydrolysate rather than reconstituted single toxins. The recalcitrant-biomass condition as a process actually presents it.	Journal SI
47	**Endo 2008 (vanillin tolerance)** Endo A, Nakamura T, Ando A, Tokuyasu K, Shima J. Appl Environ Microbiol 2008;74:7175-7185. doi:10.1128/AEM.01541-08	Vanillin is the one major hydrolysate phenolic with no dedicated screen in the built set; it is currently only folded into broad compound panels.	AEM SI (PMC2375868)
48	**Sharifpoor 2012 (kinome synthetic dosage lethality)** Sharifpoor S, van Dyk D, Costanzo M, et al., Boone C, Andrews BJ. Genome Res 2012;22:791-801. doi:10.1101/gr.129213.111	Extends the interaction paradigm into overexpression-based dosage lethality on kinases, which are the regulatory nodes flux redirection actually targets.	Genome Research SI
49	**Ambroset 2014 (metabolite QTL)** Ambroset C, et al., Fay JC. PLoS Genet 2014;10:e1004142. doi:10.1371/journal.pgen.1004142	The founding metabolite-QTL panel. Pairs with Cooper and Mulleder as the natural-variation view of a phenotype space the deletion collection covers by engineering.	PLOS Genetics SI
50	**Blank 2005 (13C metabolic flux)** Blank LM, Kuepfer L, Sauer U. Genome Biol 2005;6:R49. doi:10.1186/gb-2005-6-6-r49	The only real flux measurement tied to single-gene deletions. Everything else in the built set is a concentration or a titer, which is a proxy for flux rather than flux.	Genome Biology open-access SI
51	**Trikka 2015 (carotenogenic heterozygous screen)** Trikka FA, Nikolaidis A, Athanasakoglou A, et al., Kampranis SC, Makris AM. Microb Cell Fact 2015;14:60. doi:10.1186/s12934-015-0246-0	The second carotenoid deletion-collection screen, and the only haploinsufficiency design here. Halving dosage finds genes a null would kill, which is where the isoprenoid flux control sits. Currently listed as figure-only, so confirm Additional file 1 carries per-strain scores.	BMC Additional files 1-2
52	**Chang 2012 (organic-acid production screen)** Chang HJ, Suga H, et al. 2012. PMID 22277779. PMID 22277779	A whole-collection screen whose label is production rather than fitness. Coarse and PDF-only, but organic-acid overproduction at collection scale exists nowhere else.	PDF tables only; manual transcription required
53	**Mukherjee 2021 (CRISPRi essential genes x acetic acid)** Mukherjee V, Lind U, St Onge RP, Blomberg A, Nystrom T. mSystems 2021;6:e00410-21. doi:10.1128/mSystems.00410-21	Acetic acid is the dominant hydrolysate stressor, and this reaches the essential genes a deletion collection structurally cannot.	mSystems open-access SI

Table 3, continued

#	Dataset, citation and link	Why	Data
54	**Breslow 2008 (DAmP hypomorphic collection)** Breslow DK, Cameron DM, Collins SR, et al., Weissman JS. Nat Methods 2008;5:711-718. doi:10.1038/nmeth.1234	Partial loss of function for essential genes at the native locus, about 82 percent of the essential genome. The built set has no essential-gene hypomorph axis at all.	Nat Methods SI; Horizon YSC5050/5090/5093/5094
55	**Kofoed 2015 (barcoded ts alleles)** Kofoed M, Milbury KL, Chiang JH, et al., Hieter P, Stirling PC. G3 2015;5:1879-1887. doi:10.1534/g3.115.019174	Conditional essential-gene loss of function as point mutants at native loci, barcoded so it runs through the same pooled pipeline as the deletion collection.	G3 open-access SI; strains via Dharmacon/Horizon
56	**Lian 2017 (CRISPR-AID, beta-carotene)** Lian J, HamediRad M, Hu S, Zhao H. Nat Commun 2017;8:1688. doi:10.1038/s41467-017-01695-x	The tri-functional chassis MAGIC was built on, read out on beta-carotene. Small, but it is a combinatorial genotype with a measured product titer, which is the exact record type inverse strain design needs.	Nat Commun SI
57	**Cubillos 2017 (nitrogen-consumption QTL)** Cubillos FA, Brice C, Molinet J, et al., Martinez C. G3 2017;7:1693-1705. doi:10.1534/g3.117.042127	One of the few QTL panels whose genotype location is confirmed rather than assumed, and nitrogen assimilation sets higher-alcohol and ester formation.	BioProject PRJNA379146 (RNA-seq); genotypes in Cubillos 2013 Table S1
58	**Braberg 2020 (point-mutant E-MAP)** Braberg H, et al., Krogan NJ. Science 2020;370:eaaz4910. doi:10.1126/science.aaz4910	Sub-gene resolution genetic interactions. Narrow in gene scope, but it is the only demonstration that the interaction formalism extends below the ORF.	Science SI (accession unconfirmed)
59	**Weill 2018 (SWAT libraries)** Weill U, Yofe I, Sass E, et al., Schuldiner M. Nat Methods 2018;15:617-622. doi:10.1038/s41592-018-0044-9	The library the degron rows are built from. Ingest for its abundance and localization tables; its main value is as the cassette definition those rows' sequences resolve against.	Nat Methods SI; strains via EUROSCARF
60	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)** Nadal-Ribelles M, Islam S, Wei W, et al., Posas F, Steinmetz LM. Nat Microbiol 2019;4:683-692. doi:10.1038/s41564-018-0346-9	Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.	Nat Microbiol SI /GEO
61	**Gasch 2017 (single-cell stress heterogeneity)** Gasch AP, Yu FB, Hose J, et al., Quake SR. PLoS Biol 2017;15:e2004050. doi:10.1371/journal.pbio.2004050	Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.	PLoS Biology SI /GEO
62	**Valenti 2025 (SWAT degron + GFP collection)** Valenti M, et al. J Cell Biol 2025;224:e202409050. doi:10.1083/jcb.202409050	Depletion and its abundance consequence measured in the same strain, so enzyme dosage and its phenotype are not joined across two experiments.	J Cell Biol SI
63	**Ghaemmaghami 2003 (absolute protein abundance)** Ghaemmaghami S, Huh WK, Bower K, et al., Weissman JS. Nature 2003;425:737-741. doi:10.1038/nature02046	Converts every relative proteomics row here into molecules per cell. Without an absolute anchor, enzyme-cost and flux-capacity arguments have no units.	Nature SI (Excel)
64	**Huh 2003 (GFP localization atlas)** Huh WK, Falvo JV, Gerke LC, et al., O'Shea EK. Nature 2003;425:686-691. doi:10.1038/nature02026	The wild-type localization baseline every perturbation-imaging dataset measures change against. Compartment identity is a prerequisite for compartmentalized pathway engineering.	SGD /YeastGFP mirror

Table 3, continued

#	Dataset, citation and link	Why	Data
65	**Michaelis 2023 (protein interactome)** Michaelis AC, et al., Mann M. Nature 2023;624:192-200. doi:10.1038/s41586-023-06739-5	A physical-interaction layer to constrain which genetic interactions reflect complex membership. A graph prior rather than a phenotype.	yeast-interactome.biochem.mpg.de + ProteomeXchange
66	**Tengolics 2024 (domestication metabolome)** Tengolics R, Szappanos B, Mulleder M, et al., Ralser M, Papp B. PNAS 2024;121:e2313354121. doi:10.1073/pnas.2313354121	Shows which metabolic traits already moved under domestication, which is a prior on which are tractable to engineer further. Small n.	PNAS Dataset S11
67	**Schulte 2023 (mitochondrial complexome)** Schulte U, et al. Nature 2023;614:153-159. doi:10.1038/s41586-022-05641-w	Mitochondrial compartmentalization is a live strategy for sequestering toxic intermediates, and this maps the import and assembly bottlenecks that would cap it.	complexomics.org + ProteomeXchange
68	**Eder 2020 (flux QTL)** Eder M, Nidelet T, Sanchez I, Camarasa C, Legras JL, Dequin S. PMID 32034164. PMID 32034164	Flux as a QTL phenotype, with PDB1 and VID30 validated. The label is modeled rather than measured, which caps how far it can be trusted.	Journal SI (venue and accession unconfirmed)
69	**Yue 2017 (wild vs domesticated assemblies)** Yue JX, Li J, Aigrain L, et al., Liti G. Nat Genet 2017;49:913-924. doi:10.1038/ng.3847	Structural variation short reads miss. Relevant because industrial strains differ from S288C structurally, not only at SNPs.	ENA/GenBank (accession unconfirmed)
70	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq)** Brettner L, et al. Yeast 2024. doi:10.1002/yea.3927. doi:10.1002/yea.3927	The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.	Yeast (Wiley) SI
71	**Mulleder 2012 (prototrophic deletion collection)** Mulleder M, Capuano F, Pir P, et al., Ralser M. Nat Biotechnol 2012;30:1176-1178. doi:10.1038/nbt.2442	The background the already-built Mulleder 2016 and Messner 2023 were measured in. Auxotrophy markers distort metabolite pools, so recording which rows sit on this background is a correctness requirement, not a nicety.	EUROSCARF; Addgene kit 40276
72	**Zackrisson 2016 (Scan-o-matic growth curves)** Zackrisson M, Hallin J, Ottosson LG, et al., Warringer J, Blomberg A. G3 2016;6:3003-3014. doi:10.1534/g3.116.032342	Time-series rather than endpoint fitness. A platform, so a specific released condition panel has to be chosen before this becomes a row rather than a plan.	Open-source platform; per-study datasets
73	**Bennett 2001 (ionizing-radiation resistance)** Bennett CB, et al. Nat Genet 2001;29:426-434. doi:10.1038/ng778	Genome-stability tolerance. Only indirectly relevant, through strain robustness over long fed-batch runs where instability erodes titer.	Nat Genet SI (likely PDF)
74	**Sugiyama (lactic-acid tolerance)** Sugiyama M, Kaneko Y, et al. National Research Institute of Brewing technical report. nisr.or.jp/wp-content/uploads/NISR06sugiyama.pdf	Lactic acid is a major platform chemical, but this is a non-peer-reviewed technical report with PDF-only tables. Lowest provenance confidence in the table; treat as a hypothesis source.	NRIB technical report PDF

3 What the recommended 51 add up to ✗

Table 4: Candidates by class, split at the cut line. *Genotypes* and *Instances* sum the per-row axes; rows with no count contribute nothing, so both totals are lower bounds.

Class	In top 51	Reserve	Genotypes	Instances
Natural variation	10	4	324,786	2.8×10^6
Tolerance / robustness	14	3	90,705	5.4×10^7
CRISPR library screen	7	1	287,294	3.5×10^5
Expression / single cell	4	3	4,505	4.8×10^3
Metabolite / precursor	10	1	45,199	1.4×10^5
Modality / backbone	6	11	766,412	1.3×10^6
Total	51	23	1,518,901	5.9×10^7

The recommended set carries roughly 5.8×10^7 genotype-by-environment instances and 9.0×10^7 individual measurements. Both are lower bounds: rows whose axes could not be pinned down contribute nothing to either total, and the instance figures marked † are products of the two axes rather than counts the papers report.

Two numbers in that total deserve a discount before they are quoted anywhere, and together they are most of it. Turco 2023 supplies 3.8×10^7 of the instances and is a meta-aggregation of screens this table lists separately, so its net-new contribution is smaller than its row and currently unmeasured. Lee 2014 supplies 1.3×10^7 and is a single primary experiment, so the figure is sound, but its data are not currently reachable (Sec. 6.1). Setting both aside leaves roughly 7×10^6 instances that are both net new and in hand.

4 The six classes ✗

4.1 Natural isolates, and where sequence diversity crosses over ✗

Ten of the recommended 51 are natural-variation rows, and the crossover the class was asked for is concentrated in three of them.

Dutta 2026 (row 8) barcodes 520 natural isolates and profiles them against more than 600 compounds. Those isolates are drawn from the sequenced 1,011 panel, which is the same panel behind the already-built Caudal 2024 pan-transcriptome and behind Muenzner 2024 (row 3). One genotype axis therefore carries a genome, a transcriptome, a proteome, and now a chemogenomic response surface, which is the join that makes a transfer question askable at all.

Hale 2024 (row 4) is the sharpest instrument for that question. It crosses 8,046 CRISPRi guides over 1,721 genes into 169 sequenced segregants, so it measures directly how much a perturbation's effect depends on the background it is made in. Galardini 2019 (row 5) asks the same question with deletions instead of knockdowns, building 3,786 knockouts independently in four sequenced backgrounds across 38 conditions, and reports that about 18.5 percent of deletion phenotypes change between backgrounds.

That 18.5 percent is the reason this class matters beyond breadth. Nearly every fitness and metabolite dataset in the built set is S288C, and a model trained only there has no way to know which of its predictions survive a change of chassis.

4.2 Tolerance on recalcitrant biomass, and product tolerance ✗

Fourteen recommended rows, the largest class. Lignocellulosic inhibitor coverage in the built set currently runs through broad compound panels rather than dedicated screens, so three rows close named gaps: Pereira 2014 (row 46) screens the deletion collection against real industrial wheat-straw hydrolysate rather than reconstituted single toxins, Endo 2008 (row 47) covers vanillin, and Xiao 2014 (row 36) covers furfural through a knockdown axis that reaches essential genes a deletion collection cannot.

Product tolerance is answered mainly on isobutanol, and the earlier assumption that no genome-wide isobutanol screen exists is wrong. Kuroda 2019 (row 18) screens the deletion collection for isobutanol-specific fitness against an ethanol comparator, so the alcohol-general stress response subtracts out; its *GLN3* hit raised production 4.9-fold, which is a rare case of a tolerance screen translating into a titer. Liu 2021 (row 17) is an independent full-collection screen of the same phenotype with a different readout, giving a same-genotype, same-phenotype, different-method replicate. Crook

2016 (row 37) adds a dose-graded knockdown axis over isobutanol and 1-butanol together, and Peeters 2021 (row 13) supplies isobutanol titer as a mapped trait across 1,125 sequenced segregants.

Van Leeuwen 2024 (row 14) runs acetic, butyric and octanoic acid on one strain panel, which is what lets shared and acid-specific mechanisms separate inside a single study instead of across screens that cannot be compared.

4.3 CRISPR library screens ✗

Seven recommended rows. The already-built Lian 2019 MAGIC screen reads out furfural tolerance by growth selection; Dong 2021 (row 28) is the sibling that swaps the growth selection for an S-adenosyl-methionine biosensor and sorts by fluorescence, so the label becomes product concentration rather than fitness. That is the pattern worth copying for every precursor in Sec. 4.5, and it is the biosensor MAGIC screen the scoping request asked after.

Bao 2018 (row 32) and Roy 2018 (row 40) both move the genotype axis below the ORF, to designed single-nucleotide and small-indel variants. Both are designed rather than per-strain-verified, so both would enter as designed genotypes with a verification status. McGlincy 2021 (row 31) and Momen-Roknabadi 2020 (row 29) are two independently designed CRISPRi libraries; ingesting both makes library-to-library transfer a cheap and decisive test of whether a model has learned gene function or guide-design idiosyncrasy.

4.4 Expression across conditions ✗

Four recommended rows, and the class is genuinely thin in a way worth stating rather than papering over. Nadal-Ribelles 2025 is the largest genotype-indexed expression dataset in yeast at roughly 3,500 knockouts under osmotic stress, and it is already built. The candidates here supply what it lacks rather than more of it: Gasch 2000 (row 23) is the canonical wild-type stress-response baseline that any genotype-indexed transcriptome has to be read against, N'Guessan 2025 (row 21) profiles roughly 4,500 sequenced segregants by single-cell RNA-seq, and Leutert 2023 (row 25) covers 101 conditions at the phosphosite level, where enzyme regulation acts faster than transcription.

The initial reading was that datasets in the microbe-perturb-seq collection carry little data over large genotype numbers. That holds. Nothing found in this pass changes it, and the gap it points to is addressed in Sec. 5 rather than by another row.

4.5 Metabolite and precursor coverage ✗

Ten recommended rows. The precursor accounting in `notes/metabolism.central-carbon-precursors.md` sets the target: of the twelve central carbon precursors, acetyl-CoA is measured on 19 strains and succinyl-CoA and malonyl-CoA on none, so identity coverage is good and statistical power is not.

Blank 2005 (row 50) is the only dataset in this table that measures flux rather than a concentration or a titer, across roughly 200 deletion mutants, and it is the only real handle on acetyl-CoA turnover under genetic perturbation. Malonyl-CoA still has no direct measurement anywhere; the nearest handles remain its dominant sinks, which the already-built Xue 2025 free fatty acids and da Silveira 2014 lipidomics supply, and Zhu 2014 (row 35) extends with a lipidome across 129 kinase and phosphatase deletions.

Cooper 2010 (row 9) is the highest-value row in the class for a reason unrelated to novelty. It measures the same amino-acid pools as the already-built Mulleder 2016, on an overlapping genotype axis, by capillary electrophoresis instead of mass spectrometry. Two independent platforms measuring one trait class is the cleanest available test of whether a model has learned biology or a batch effect, and nothing else in the built set offers it.

On carotenoids, Ozaydin 2013 is built. Trikka 2015 (row 51) is the other high-throughput carotenoid screen: 4,700 heterozygous deletions scored by color, then carried through iterative rounds to a sclareol titer. Halving gene dosage rather than deleting outright is what finds the isoprenoid flux control points a null would kill, and no other row in this table uses that design. It is currently recorded as figure-only, so confirming that Additional file 1 carries per-strain scores is a precondition, not a formality.

4.6 Perturbation modality and reference backbone ✗

Six recommended rows, none of which is a headline dataset and one of which is load-bearing. Puddu 2019 (row 19) is whole-genome sequence for every strain in the deletion collection. Twenty-one recommended rows declare an **S288C-KO** sequence basis, and without Puddu that basis is an assumption that each strain differs from the reference at exactly the intended locus. Puddu shows it frequently does not. Ingesting it converts a shared assumption across half this table into a measurement, which is worth more than its own 4,800 instances suggest.

5 Rows that bear on a yeast Perturb-seq ✗

Eight rows bear on whether to run a genome-scale Perturb-seq in yeast, five of them inside the recommended 51. Scale ranking buries three of the eight, which is the correct ranking and the wrong emphasis: a method paper that measures the within-clone transcript noise floor has an instance count of two and settles whether the experiment is worth running at all.

Table 5: Rows bearing on a yeast Perturb-seq, listed regardless of rank. *Rank* is the row's position in Table 2.

Rank	Dataset	What it settles
4	**Hale 2024 (CRISPRi x natural variation)**	Directly measures how a genetic perturbation's effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.
21	**N'Guessan 2025 (segregant scRNA-seq eQTL)**	Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale.
27	**Chong 2015 / CYCLoPs (single-cell proteome dynamics)**	Per-cell rather than per-strain-mean labels, over 20 million cells. The distributional readout a Perturb-seq design has to be scored against.
29	**Momen-Roknabadi 2020 (inducible CRISPRi library)**	A second CRISPRi library on a different vector and guide-design rule. Whether a model trained on one library transfers to the other is a cheap, decisive test of guide-level overfitting.
31	**McGlincy 2021 (genome-scale CRISPRi library)**	Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.
60	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)**	Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.
61	**Gasch 2017 (single-cell stress heterogeneity)**	Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.
70	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq)**	The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.

Three of these answer distinct design questions rather than supplying training data. Brettner 2024 is the chemistry the costing model in `experiments/024-perturb-seq-costing/` is built on. Nadal-Ribelles 2019 measures within-clone transcript correlation, which is the floor any per-guide mean has to clear and therefore what sizes the cells-per-guide budget. Gasch 2017 separates intrinsic from extrinsic expression heterogeneity under stress, which sets what fraction of the variance in a pooled screen is recoverable signal rather than cell state.

The two that would most change the design are already ranked high. McGlincy 2021 (row 31) is the most plausible library to build on, since ten guides per gene gives a graded rather than binary knockdown and it reaches essential genes. N'Guessan 2025 (row 21) is the closest existing analog to the proposed experiment, pairing per-cell expression with sequenced genotypes at scale, though its genotypes are recombinant natural variants rather than engineered knockdowns.

5.1 The synergy question is answerable, and not yet answered ✗

Whether one dataset predicts another is the question that decides how much new data is worth collecting, and this list contains the pairs needed to test it before any is collected:

- **Same phenotype, different platform.** Cooper 2010 against the built Mulleder 2016, amino acids by capillary electrophoresis against mass spectrometry on an overlapping genotype axis.
- **Same phenotype, different method.** Liu 2021 against Kuroda 2019, isobutanol tolerance on the same collection by colony imaging against competitive fitness.

- **Same perturbation, different background.** Hale 2024 and Galardini 2019, both of which vary the background while holding the perturbation fixed.
- **Different library, same modality.** McGlincy 2021 against Momen-Roknabadi 2020, two CRISPRi libraries with different vectors and guide-design rules.
- **Cheap layer against expensive layer.** Caudal 2024 transcriptomes against Muenzner 2024 proteomes, on overlapping isolates from the 1,011 panel.

None of these transfers has been measured. Each is a hypothesis about what a model could learn, and each becomes a run once both halves of the pair are ingested. The last pair is the one that bears most directly on the Perturb-seq costing, since it asks whether a cheap expression readout carries the information an expensive one would.

6 What was dropped ✗

Table 6: Considered and dropped. *no-sequence* is the hard gate: without a route to the strain's genomic content there is no genotype to map a phenotype from.

Dataset or group	Rule	Reason
Snoek 2015 robot-assisted genome shuffling (ethanol tolerance)	no-sequence	Shuffled progeny are recombinants of unrecorded parental segments with no per-strain sequencing, so the genotype cannot be reconstructed.
Adaptive laboratory evolution / EMS mutagenesis tolerance panels (e.g. farnesol-FPP selection)	no-sequence	Mutations are unmapped, so a strain's genome is unknown even in principle without whole-genome resequencing the study did not do.
Avalos lab isobutanol-biosensor deletion screen	not-a-dataset	Described only in a PhD thesis and a DOE report. Track for publication; not citable as a dataset.
Kitamoto/Kaneko ester and higher-alcohol screen	not-a-dataset	Citation could not be pinned to a specific paper. A placeholder, not a located dataset.
Fuhrer/Zamboni flow-injection metabolomics	off-species	The validated genome-scale application is in E. coli. No yeast deletion-collection version was located.
Anglada-Girotto 2022 (CRISPRi + metabolomics)	off-species	352 CRISPRi genes against 1,342 drug-induced metabolic changes, but in E. coli. Verified from the mirrored PDF.
Ozturk 2022 proteome-effects screen	off-species	3,308 deletions in S. pombe. Out of species scope.
Chen 2024 K. phaffii CRISPR fitness screen; Y1000+ multi-species panel	off-species	Not S. cerevisiae. Revisit if the generalization axis widens to other hosts.
iIsor850 genome-scale model (Issatchenkia orientalis)	not-a-dataset	A reconstruction, not an experiment, and off species. No genome-wide I. orientalis phenotype screen was located; the CABBI knockout library is unpublished.
CABBI 13C-MFA kinetic-model repository	not-a-dataset	16 knockout strains total, and the released fluxes are K-FIT model fits recapitulating 75-77 percent of measurements, not raw measurement. Blank 2005 supersedes it on both counts.
Sc2.0 / Sc3.0 synthetic-genome design	not-a-dataset	Genome-engineering programs, not gene-indexed perturbation-phenotype data.
Lee 2014 / Hoepfner 2014 / Vanacloig-Pedros 2022 / Messner 2023 / Mulleder 2016 / Lian 2019 / Mormino 2022 / Nadal-Ribelles 2025 and 12 others	already-built	Named as top candidates by the 2026-07 triage pass and BUILT since. Only Lee 2014 remains outstanding and is kept as row 1.

The **already-built** row in that table is the largest single correction this pass makes. A July 2026 triage pass named a top ten for immediate ingestion; nine of those ten have since been built, along with roughly a dozen other rows it ranked highly. Only Lee 2014 remains outstanding, and it is row 1 here. Any list that still treats that top ten as outstanding work is stale.

6.1 Two rows are not untouched candidates ✗

Checking a name against the built list is not sufficient, because a dataset can carry a loader, or a failed retrieval attempt, without appearing there. Two rows do.

Lee 2014, row 1, is blocked rather than available. WS15 already attempted this ingestion. The roadmap records it as awaiting author matrices, which means the Science supplementary tables and the Nislow and Giaever lab portal did not yield per-strain values. Its rank is still earned: 3,356 compounds against the full heterozygous and homozygous collections really is the largest chemical-genetic matrix in existence, and it really is on the YKO genotype axis the built set already uses. What changes is the nature of the work. Row 1 is an author request, not a loader, and the 1.3×10^7 instances it contributes to the totals in Sec. 3 should not be counted as reachable until that request succeeds.

Turco 2023, row 20, already has a loader. `torchcell/datasets/scerevisiae/yeastphenome.py` consumes the curated YeastPhenome release and retains 49 growth screens, so this row is in flight rather than net new. It is not in the built set, which is why it appears here at all. Two consequences follow. Its de-duplication problem is partly solved already, since the loader excludes the primaries built directly, along with heterozygous-diploid and expression screens. And Lee 2014 is not among its 49 screens, so ingesting YeastPhenome does not backfill row 1.

7 Before a loader is written ✗

Every figure in Table 2 is an axis count, not a verified record count, and several are weaker than that. Three qualifications apply, in decreasing order of how much damage they would do if ignored.

Instance counts marked † are products, not counts. A screen reported as 4,800 strains against 3 conditions is entered as 14,400 instances. Real screens drop strains per condition, so each such figure is an upper bound. The ones marked ‡ are weaker still: order-of-magnitude estimates for libraries whose exact size was not confirmed this pass. Neither should be quoted as a dataset size without opening the source.

Phenotype dimensionality is estimated where no fixed panel is published. Ranking uses instances times dimensionality, and dimensionality is exact for a stated panel such as Ambroset 2014 at 74 metabolites, but estimated for open-ended ones. Proteome rows are entered at 2,000 proteins per strain and transcriptome rows at 6,000 genes. Those estimates move rank order, so a row whose position looks surprising should have its dimensionality checked first.

Accessions marked unconfirmed were not fetched. The accession column separates locations verified live this pass from those taken from a paper's data-availability statement. Confirmed live: GEO GSE123118 with its companion GitHub release for Galardini 2019, SRA PRJNA986287 for Hale 2024, PRIDE PXD048219 for Muenzner 2024, Dryad for Chica 2026, ChemGRID for Wildenhain 2016, and BioProject PRJNA379146 for Cubillos 2017. Everything else is a claim from a publication, and the pattern is well established that a data-availability statement and a reachable file are different things.

Four rows carry a risk specific enough to name. Trikka 2015 is recorded as figure-only, so whether Additional file 1 holds per-strain scores decides whether row 51 is ingestible at all. Chang 2012 and the Gurvitz fatty-acid screen are PDF-table-only and need manual transcription. Sugiyama is a non-peer-reviewed technical report and has the lowest provenance confidence of anything listed; it belongs in the reserve as a source of candidate genes rather than as training data.

Two rows are not single datasets. Turco 2023 is a meta-aggregation and Zackrisson 2016 is a platform. Neither reduces to one loader: Turco needs a de-duplication pass against the rest of this table before its net-new content is known, and Scan-o-matic needs a specific released condition panel chosen before it becomes a row rather than a plan.